MIL-I-46877(MI)
14 December 1973
SUPERSEDING
MIS-11266
14 December 1970

MILITARY SPECIFICATION

INSULATING COMPOUND, ELECTRICAL

This specification is approved for use by all activities of the Department of the Army.

1. SCOPE

1.1 Scope. This specification covers the requirements for a two part, solvent free, clear, flexible silicone resin system for use in packaging electronic equipment.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

VV-I-530	Insulating Oil, Electrical (for Transformers, Switches, and Circuit Breakers)

Military

MIL-I-16923	Insulating Compound, Electrical, Embedding.

STANDARDS

Federal

FED-STD-141	Paint, Varnish, Lacquer and

FSC 5970

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	Related Materials, Methods of Inspection, Sampling and Testing.
FED-STD-406	Plastics, Methods of Testing.
FED-STD-601	Rubber, Sampling and Testing.

Military

MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing Materials

ASTM D 570	Apparent Density, Bulk Factor and Pourability of Plastic Materials, Method of Test for

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Qualification. Material furnished under this specification shall be a product which has been tested and passed the qualification tests specified herein.

3.2 Material.

3.2.1 Color. Clear.

3.2.2 Viscosity (uncured). 800 - 1500 centipoises.

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3.2.3 Hardness. 12 durometer (Shore A) minimum.

3.2.4 Shrinkage. 1 percent maximum.

3.2.5 Tensile strength. 35 plus or minus 10 pounds per square inch (psi).

3.2.6 Elongation. 100.0 plus or minus 30.0 percent.

3.2.7 Moisture absorption. 0.1 percent increase in weight maximum.

3.2.8 Water absorption. 0.1 percent increase in weight maximum.

3.2.9 Dielectric strength. 450 volts/mil minimum.

3.2.10 Dielectric constant. 3.00 maximum.

3.2.11 Dissipation factor. 0.002 maximum.

3.2.12 Insulation resistance. 2×10^{12} ohms minimum.

3.2.13 Deformation under load. 5 percent maximum.

3.2.14 Mix ratio and cure. Mix ratio shall be 0.5 parts by weight curing agent per 100 parts by weight of silicone compound. Curing shall be 4 hours at 150° Fahrenheit (F).

3.2.15 Shelf life (Uncured). 6 months minimum from date shipped, stored at room temperature.

3.3 Workmanship. The materials furnished under this specification shall be made to acceptable industry standards. There shall be no contamination or inclusions.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of tests. Inspection and testing shall be classified as follows:

a. Qualification Tests.

b. Quality Conformance Tests.

4.2.1 Qualification tests. Qualification tests shall consist of tests for all requirements of this specification.

4.2.2 Quality conformance tests. Quality conformance tests shall consist of tests for the following requirements:

a. Color.

b. Viscosity (Uncured).

c. Hardness.

d. Dielectric Constant.

e. Dissipation Factor.

f. Mix Ratio and Cure.

4.3 Test methods.

4.3.1 Color. Visual examination.

4.3.2 Viscosity. Viscosity shall be determined in accordance with FED-STD-141, Method 4287.

4.3.3 Hardness. Hardness shall be measured in accordance with FED-STD-601, Method 3021.

4.3.4 Shrinkage. Shrinkage shall be determined as follows: The volume of a cubic mold, approximately 1.0 inch on each side, shall be determined at 77° plus or minus 2° F. The test material shall be mixed, poured into the mold and cured according to paragraph 3.2.14. The material shall then be removed from mold, examined and the volume determined by water displacement method at 77° plus or minus 2° F. The percent shrinkage shall be calculated by

$$\text{Percent Shrinkage} = \frac{V_1 - V_2}{V_1} \times 100$$

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where V_1 = volume of mold

and V_2 = volume of material.

4.3.5 Tensile strength. The tensile strength shall be determined in accordance with FED-STD-601, Method 4111.

4.3.6 Elongation. The elongation shall be determined in accordance with FED-STD-601, Method 4121.

4.3.7 Moisture absorption. Moisture absorption shall be determined per moisture absorption paragraph in MIL-I-16923 .

4.3.8 Water absorption. Water absorption shall be determined in accordance with ASTM-D-570.

4.3.9 Dielectric strength. Dielectric strength shall be determined in accordance with FED-STD-406, Method 4031 using 0.063 inch thick specimen, 0.250 inch electrodes, under insulating oil per VV-I-530 A.

4.3.10 Dielectric constant. The dielectric constant shall be determined in accordance with FED-STD-406, Method 4021.

4.3.11 Dissipation factor. The dissipation factor shall be determined in accordance with FED-STD-406, Method 4021 at 1 KHz and 77° plus or minus 2° F.

4.3.12 Insulation resistance. Insulation resistance shall be determined by preparing five specimens 0.44 inch high x 1.50 inches in diameter from test material. Each specimen shall have two brass electrodes, 0.125 inch diameter x 1.75 inches long, cast in the center of the 1.50 inches diameter - 0.125 inch apart and 0.010 inch from the base of the sample. After curing seal surface around electrodes with silicone grease, check resistance of specimens using a Model 1620C Freed Megohmeter or equivalent. Soak specimens in distilled water at 110° to 118° F. for 30 days. Check resistance after returning samples to room temperature and drying in air blast.

4.3.13 Deformation under load. Deformation under load shall be determined by preparing ten cubes, approximately 1 inch on each side, from test material. Measure specimens with a micrometer. Load five specimens with 5 pounds each at 77° plus or minus 2° F. for 10 days. Load five other specimens with 5 pounds each at 165° plus or minus 2° F. for 10 days. At the end of 10 days measure specimens at 77° plus or minus 2° F. for percent deformation.

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4.3.14 Mix ratio and cure. Determine by preparing a 100 gram batch in accordance with 3.1.14 and submitting to visual inspection.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be level A, B, or C as specified in accordance with Section 5 of MIL-I-16923.

5.2 Marking. Each container shall be marked in accordance with MIL-STD-129, The Precautionary Markings, and shall also include the following:

a. Specification number and lot number.

b. Manufacturer's name.

c. Purchase order number and product designation.

d. Date of product shipment.

e. Selection of applicable level of preservation, packaging, and packing.

6. NOTES

6.1 Ordering data. Procurement documents should specify the title, number and date of this specification including revision level, if applicable.

6.2 Supersession data. This specification includes the requirements of MIS-11266, dated 14 December 1970.

Custodian:
ARMY - MI

Preparing Activity:
ARMY - MI
Project No. 5970-A425

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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